

PortFlow AI

Autonomous Maritime Terminal Operations Intelligence Platform — Predictive Congestion Management, Dynamic Berth Allocation & Grounded Operations Copilot

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Core Value Proposition

Proactive 72-hour operational intelligence replacing reactive spreadsheets with GPU-accelerated digital twins, OR-Tools optimization, and IBM Bob RAG decision synthesis.

The High Cost of Port Congestion

80% of global trade moves by sea. Marine terminals face cascading bottlenecks costing millions daily.

\$50K+

Daily Demurrage Penalty

Every day a container vessel sits idle at outer anchorage costs carriers and shippers tens of thousands in direct penalties.

4–7 Days

Average Bottleneck Dwell

Disruptions like crane failures or sudden weather surges cause compounding delays across berths, yards, and railheads.

Reactive

Legacy TOS Infrastructure

Current terminal operating systems rely on manual whiteboard scheduling and siloed shift spreadsheets with zero predictive horizon.

The PortFlow AI Solution

An end-to-end mission-control cockpit uniting prediction, optimization, and generative agent assistance.

1. Multi-Horizon Forecasting

Statistical ML models projecting quayside congestion trajectories across **6h, 12h, 24h, 48h, and 72h** with spatial hotspot risk scoring.

2. OR-Tools Optimization

Constraint programming (CP-SAT) engine resolving vessel-to-berth and crane assignments to minimize dwell and demurrage.

3. IBM Bob RAG Copilot

Specialized maritime domain AI assistant grounded in live terminal telemetry vectors to generate 1-click recovery directives.

60 FPS Canvas Digital Twin

Ultra-responsive geospatial visualization of quayside berths, gantry cranes, container blocks, and anchorage basins.

Interactive GIS Layout

Vector-drawn terminal geometry rendering Berths B1–B4, Quay Cranes QC-01–QC-08, Container Yards A–D, and outer anchorage zones with zero layout thrashing.

Includes vessel AIS search, dynamic status filtering (Waiting, Berthed, In-Transit), and slide-over entity inspector drawer.

Anti-Jank Performance Architecture

Built strictly following GPU layer promotion (`translateZ(0)`), subpixel typography antialiasing, and composite-only transitions (`opacity`, `transform`).

Instantaneous UI responsiveness even under high-density spatial entity loads.

What-If **Disruption Simulator**

Stress-test terminal resilience against unexpected crises and simulate recovery strategies in real time.

Crane Outage Stress Test

Simulates crane QC-03 mechanical breakdown. Instantly recalculates berth capacity drop and vessel queue build-up.

Storm & Influx Surges

Models +20 vessel crisis arrival spikes and weather closures. Evaluates anchorage overflow risks before ships arrive.

Automated Recovery Directives

Recommends dynamic diversion gates, overtime gang shifts, and priority vessel reallocations with delta metrics.

Measurable Impact & Deployment

Proven ROI for port terminal operators and global maritime logistics networks.

-31.6%

Waiting Time Reduction

Demonstrated average anchorage dwell reduction via OR-Tools optimized berth and crane scheduling.

\$2.4M+

Annual Demurrage Savings

Direct penalty avoidance per major terminal by preventing idle anchorage queues and vessel turnaround overlaps.

Cloud-Ready

Enterprise Architecture

Designed for multi-tenant deployment with MongoDB Atlas change streams and watsonx.ai enterprise orchestration.